Large-scale traffic flow reconstruction from heterogeneous mobility sensors

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Abstract

Regional traffic reconstruction must combine observations that differ in spatial support, sampling and independence from the model. We assess a workflow for Emilia-Romagna, Italy, in which mobile-phone origin–destination activity drives an agent-based simulation on an OpenStreetMap road graph and inductive loop detectors correct the simulated counts locally. GPS probes, kept out of the reconstruction, test the hourly profiles of all matched links in two campaigns, one before and one after the COVID-19 restrictions. Because the probe counts are rescaled by the simulated volumes, this comparison tests the temporal shape of traffic only; volumes are tested by withholding each loop detector in turn. The reconstructed profiles are only marginally closer to the GPS profiles than the injected demand profile, less accurate than a mean detector profile, and worse on weekends. Assimilating the neighbouring detectors reduces the error on a withheld detector by about three percentage points, while more than half of its observed volume remains unexplained. Part of the discrepancy is traced to an approximately one-hour simulated lead, missing late-night demand and a detector clock mismatch. The results delimit what heterogeneous regional observations support: network-wide coverage by a modelled field, weak transfer of local count information, and no demonstrated advantage over simple temporal baselines.

Keywords: traffic reconstruction, mobile-phone demand, inductive loop detectors, GPS validation, agent-based simulation, sparse observations

Working research draft, 27 October 2026. Archived implementation assessed; corrected-clock experiments and author confirmations remain outstanding.

1 Introduction

Mobile-phone records provide a geographically broad view of travel demand. Their regularities in space and time [1] have been used to estimate origin–destination (OD) matrices by time of day and trip purpose [2–4], and passively collected data are increasingly part of transport modelling practice [5–7]. Converting activity between areas into traffic on roads, however, requires assumptions about vehicle use, trip endpoints and routing, and phone-based samples carry their own ownership and activity biases [8]. Previous studies have combined mobile-phone demand with road networks and assignment models to estimate road usage and congestion [9–11], building on the equilibrium-assignment tradition [12]. Our group has studied GPS and mobile-device

mobility in Italian cities, including Bologna and Rimini in Emilia-Romagna [13–18]. The remaining question for a regional application is what can be established independently about the resulting link counts and their variation through the day.

Fixed detectors and vehicle probes address different parts of this question. Detectors measure passages at a small set of road sections and have long been used to constrain OD estimation [19–21] and to calibrate the demand of agent-based simulations [22]. GPS observations cover additional roads but sample a fleet whose inclusion probability need not be constant between links or hours [23–26]. Traffic-state estimation and multi-sensor fusion therefore depend on both the physical model and the observation process [27–32]. Sensor location and flow observability place further limits on what sparse counts can identify [33–36]. In particular, reproducing the count of a detector already used for correction is not evidence of accuracy elsewhere.

These issues remain relevant alongside recent methods. Telecom–vehicle fusion has been studied for prediction away from cameras [37]; probabilistic estimation targets sensor-free traffic states [38]; and sparse-detector scaling methods estimate network traffic states from limited stationary coverage [39]. Uncertainty in demand inputs propagates through every stage of such models [40, 41], and simulation outputs require statistical validation against data not used for calibration [42, 43]. Simple historical profiles are also strong benchmarks for temporal traffic patterns [44, 45]. The present study does not introduce a new fusion principle or establish that its algorithm outperforms those methods. It evaluates an existing regional workflow against observations withheld from demand generation and correction, with simple baselines and an explicit separation between temporal-profile and count validation.

We examine a reconstruction for Emilia-Romagna, Italy. January 2020 mobile-phone OD activity drives the `minimocas` simulator on an OpenStreetMap-derived graph. Regional inductive loop detectors, called coils below, correct the simulated link counts locally. Viasat GPS extracts from February 2020 and June–July 2021 provide a separate probe layer. Our assessment uses every matched link with positive simulated and GPS totals, a leave-one-counter-out experiment, day-type comparisons and an audit of timing and implementation choices. Clustering and selected-road examples are retained in the Supplementary Material as illustrations rather than evidence of network-wide accuracy.

2 Results and validation

2.1 Regional inputs and reconstructed field

The regional network connects an east–west urban corridor, the Adriatic coast and Apennine roads (Fig. 1). Mobile-phone activity is aggregated to municipality-level OD relations. The demand basis is January 2020 throughout the experiments; it is not a measurement of daily vehicle trips in either validation campaign. Weekday and weekend profiles differ, with more concentrated peaks on weekdays (Fig. 2). Four endpoint classes represent observed OD pairs and residual demand with one or both endpoints sampled by the model. These classes allocate mobility activity to the network under the assumptions described in Methods.

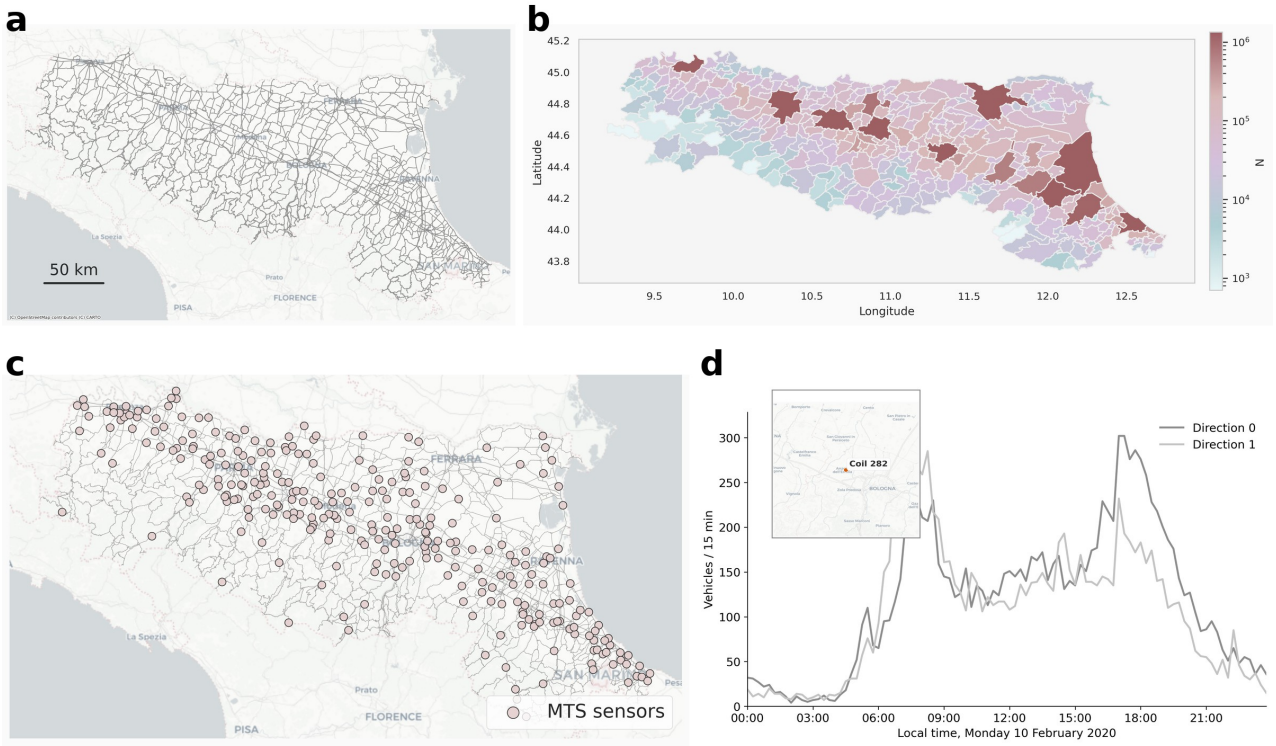


Fig. 1 Regional domain and observational inputs. (a) Model road graph on the inherited CartoDB Positron basemap; scale bar, 50 km. **(b)** Municipality-level mobile-phone OD activity in January 2020, shown logarithmically. **(c)** MTS loop-detector locations. **(d)** Directional counts of coil 282 on 10 February 2020 in local time, with its location in the inset. Panels are retained from the revised source manuscript; their complete regeneration and basemap attribution remain part of the figure-provenance audit.

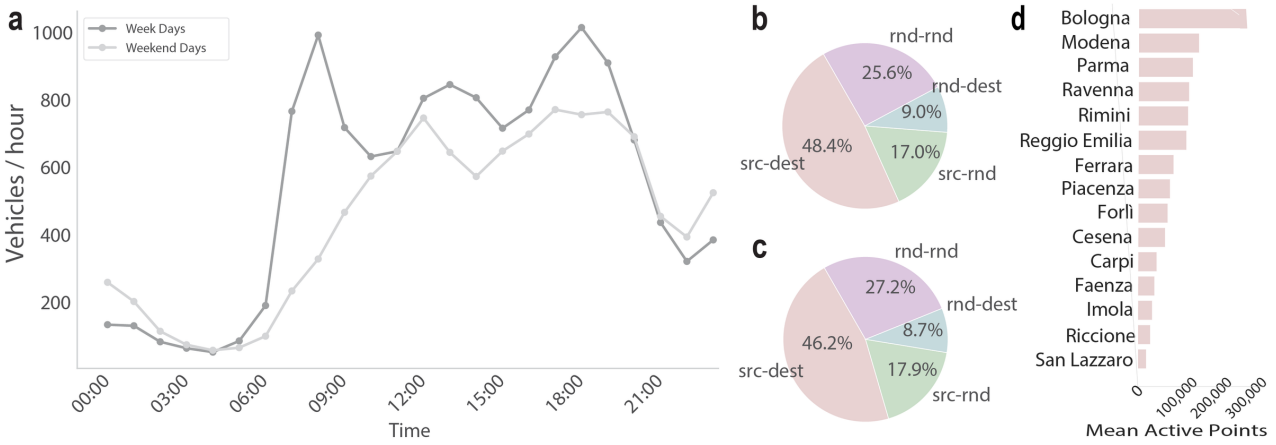


Fig. 2 Mobile-phone demand basis. (a) Mean hourly weekday and weekend activity in January 2020. **(b,c)** Weekday and weekend shares of source–destination, source–random, random–destination and random–random demand. **(d)** Activity of the 15 most active municipalities. These describe the input activity and its model decomposition, not independently measured vehicle flows. Panels are retained from the revised source manuscript.

The simulator routes agents over directed links and records distinct agents present on each link during hourly intervals. This unique-link count differs from passage counts at a point detector. On the polyline containing coil 282, the raw 10 February 2020 reconstruction carries 14,804 agents against 21,935 observed passages, with hourly correlation 0.95 in the local-time comparison (Fig. 3a; inherited illustrative result). Assimilation imposes the detector series under non-zero simulated-count conditions. Its agreement with that detector is therefore a consequence of the update. The daily map provides a regional model output, rather than a map of independently validated traffic volumes.

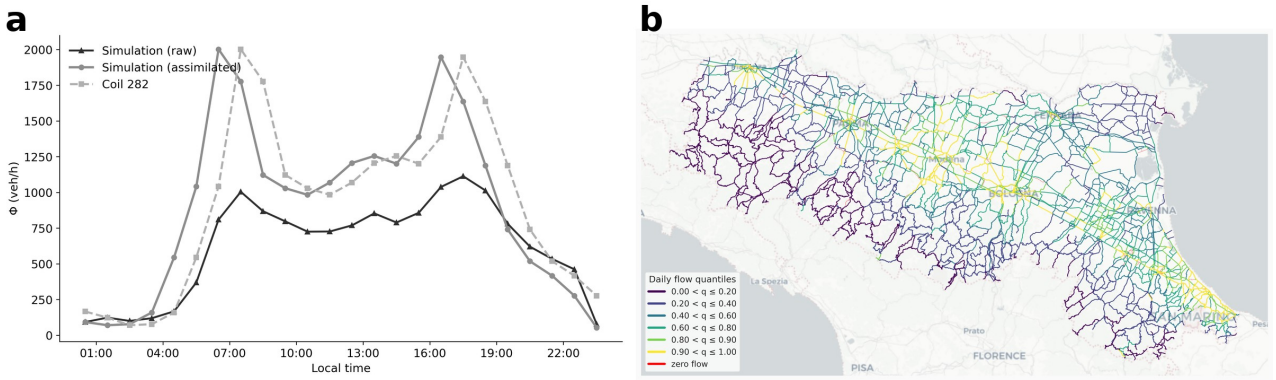


Fig. 3 Model output and an assimilated location. (a) Raw and assimilated counts on the link of coil 282 and observed hourly counts on 10 February 2020. The archived assimilation treats UTC observations as local time, explaining the relative displacement of its series. **(b)** Simulated daily counts by flow quantile. Counter agreement after assimilation is not held-out validation. Panels are retained from the revised source manuscript.

2.2 What the independent GPS comparison measures

Viasat observations were not used to generate demand or assimilate counters. Their matched links cover a subregion, predominantly near Forlì–Cesena (Fig. 4); validation there cannot establish accuracy across the entire regional graph. We estimate a link-specific GPS-to-simulation ratio over the same dates,

$$\eta_i = \frac{\sum_{d,h} \psi_i(d,h)}{\sum_{d,h} \phi_i(d,h)}, \tag{1}$$

where ψ_i counts probes and ϕ_i counts simulated agents. Rescaling GPS counts by η_i matches the simulated total by construction. Accordingly, η_i is a model-relative sampling ratio, not an independently measured penetration probability. It cannot validate absolute volumes or establish that fleet coverage is uniform across hours.

We compare normalized hourly shares,

$$p_i(h) = \frac{\sum_d \phi_i(d,h)}{\sum_{d,h'} \phi_i(d,h')}, \qquad q_i(h) = \frac{\sum_d \psi_i(d,h)}{\sum_{d,h'} \psi_i(d,h')}, \tag{2}$$

and their mean absolute difference in percentage points,

$$\mathrm{MAE}_i = \frac{100}{24} \sum_{h=0}^{23} |p_i(h) - q_i(h)|. \tag{3}$$

Dividing by the mean hourly share, 100/24, gives

$$\varepsilon_i = \sum_h |p_i(h) - q_i(h)|, \tag{4}$$

an L^1 distribution distance between zero and two, not a relative vehicle-count error. Both directions are summed before the GPS comparison, whereas the counter experiment retains directions.

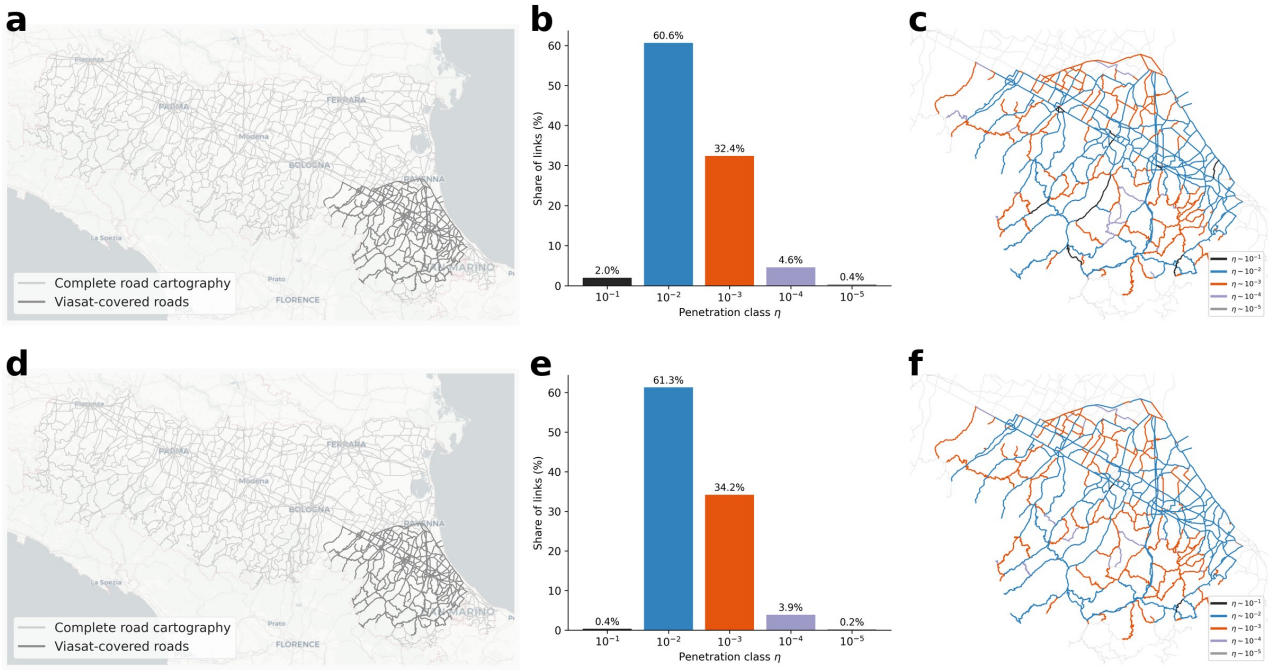


Fig. 4 GPS coverage and model-relative sampling ratios. Top: 10–17 February 2020; bottom: 28 June–4 July 2021. **(a,d)** GPS-matched links against the regional graph. **(b,e)** Link shares by η_i rounded to the nearest power of ten. **(c,f)** Spatial arrangement of these classes in the validation area. The classes depend on simulated volumes and are not independent measurements of fleet penetration. Panels are retained from the revised source manuscript.

2.3 All-link assessment and temporal baselines

Of 930 matched links in 2020 and 927 in 2021, respectively 848 and 845 have positive totals in both data sources. The campaigns contain eight dates in 2020 (six weekdays and two weekend days) and seven in 2021 (five weekdays and two weekend days). The extra archived simulation of 5 July 2021 is excluded because the GPS extract ends on 4 July. Regenerating the archived MAE, correlation and OD-baseline metrics reproduces the previous calculations to numerical precision.

Median assimilated MAE is 1.34 pp in 2020 and 1.49 pp in 2021, with interquartile ranges 1.05–2.01 and 1.15–2.12 pp. These correspond to median ε of 0.32 and 0.36. Median Pearson correlation of hourly shares is 0.82 and 0.76. Full distributions, rather than selected good-fitting roads, are shown in Fig. 5. Error is associated with the model-relative sampling class, but classes depend on simulated volume and cannot by themselves separate probe noise from reconstruction error (Supplementary Section S5).

Three common temporal profiles provide baselines (Table 1). A uniform profile ignores within-day variation. The injected OD-demand profile measures the value added by network loading. The mean normalized MTS profile supplies observed timing without link-specific assignment. The 2020 MTS profile uses the 82 detector directions complete across the validation dates, after conversion to local time. The 2021 comparator uses 2022 summer weekday-specific averages, excluding incomplete direction profiles; it is a cross-year comparator, not contemporaneous validation.

The reconstruction improves little on the OD profile: the median paired reduction in MAE is 0.063 pp in 2020 and 0.071 pp in 2021. The MTS baseline is more accurate, with median errors 1.15 and 1.22 pp. Assimilation does not improve GPS shape errors: the raw simulation gives 1.33 and 1.39 pp. These outcomes provide no evidence that local count correction improves hourly shape on the GPS-matched network.

Table 1 Median hourly-share MAE (pp) on all usable links in each campaign and day type. Link sets depend on non-zero totals within the selected days. MTS in 2021 is a 2022 seasonal comparator, excluding direction profiles with missing hours.

Campaign	Days	Links	Raw	Assimilated	OD	MTS	Uniform
2020	all (8)	848	1.33	1.34	1.40	1.15	2.70
2020	weekday (6)	844	1.53	1.55	1.63	1.32	2.86
2020	weekend (2)	840	1.96	1.96	1.96	1.88	2.96
2021	all (7)	845	1.39	1.49	1.52	1.22	2.56
2021	weekday (5)	843	1.57	1.69	1.70	1.37	2.71
2021	weekend (2)	829	2.12	2.18	2.13	2.01	2.92

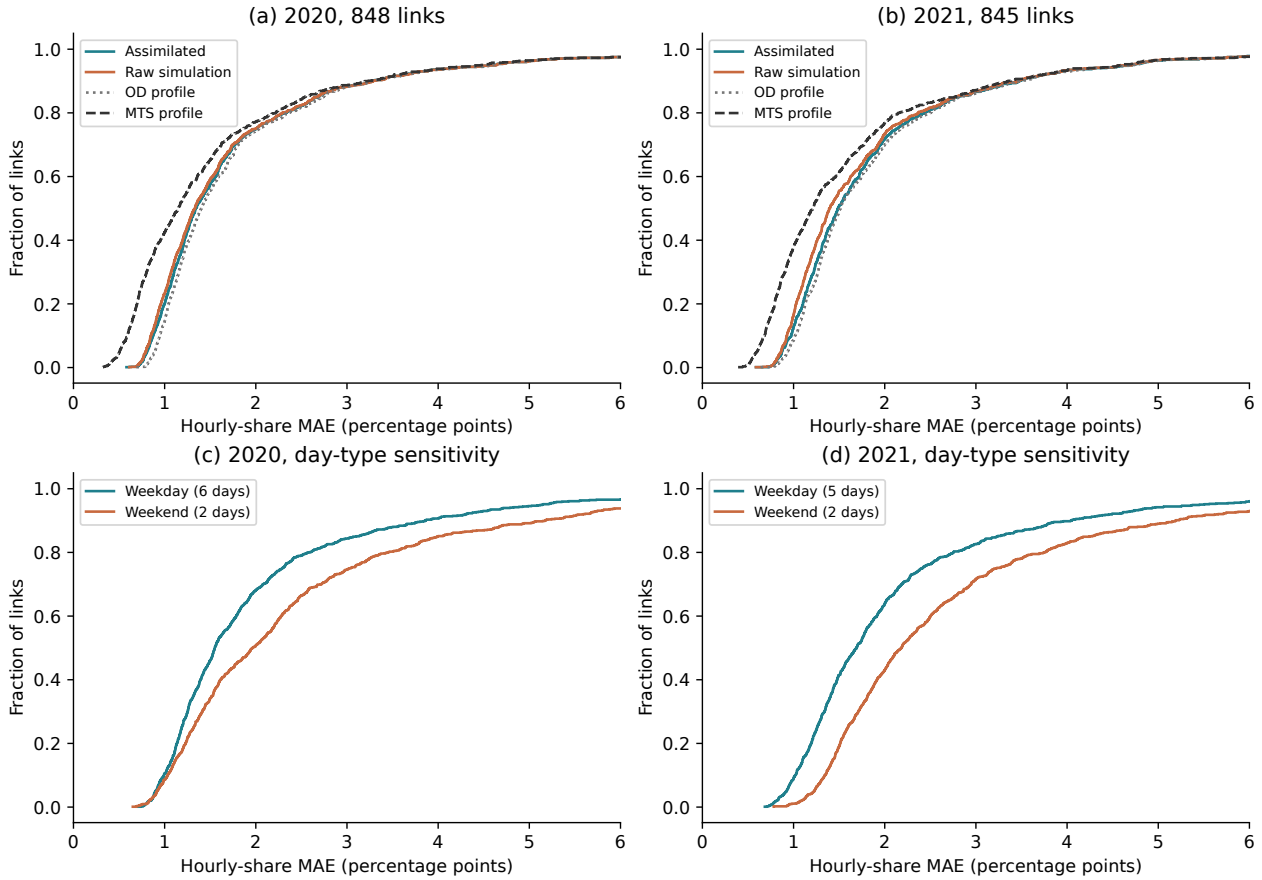


Fig. 5 Hourly-share validation on all usable matched links. (a,b) MAE distributions for raw and assimilated reconstruction, OD timing and the mean MTS profile. **(c,d)** Assimilated errors after separating weekdays and weekends; each weekend contains two days. Curves extend beyond the displayed 6 pp limit. Different day types have different usable-link sets and probe counts. The MTS comparator for 2021 comes from 2022.

2.4 Day-type and timing diagnostics

Restricting to weekdays gives median assimilated MAE 1.55 pp on 844 links in 2020 and 1.69 pp on 843 in 2021. On the two weekend dates these rise to 1.96 pp on 840 links and 2.18 pp on 829 links. MTS baseline errors remain lower, at 1.88 and 2.01 pp for weekends. The deterioration is descriptive: separating days also reduces probe totals and changes the non-zero support. It cannot be attributed solely to a worse weekend demand model.

Pooled simulated link counts peak at 17:00, one hour before pooled GPS counts in both campaigns (Fig. 6a,b). These pooled curves weight roads by each data source's counts and are diagnostic, not regional population estimates. No fitted shift is applied in the primary validation. A previous one-hour-shift sensitivity lowers median MAE to 1.29 and 1.34 pp; it is not evidence that a physical timing correction has been identified.

The configuration generator omits late-night demand, and the archived assimilation compares UTC counter timestamps with the local simulation clock. Both mechanisms require separate correction and rerun experiments.

2.5 Transfer of counter information to held-out links

The archived leave-one-counter-out test covers 149 coils and 610 coil-days in February 2020. The held-out coil is removed both from the assimilated observations and from the list of sensor links stopping propagation. The archived result contains 1,202 directional records, 18 fewer than two per coil-day; directions missing from the correction output were skipped by its runner. On the retained support, pooled absolute error is 59.8% of observed volume before assimilation and 56.7% after assimilating the other coils (Fig. 6c). The reduction is 3.08 percentage points, not a 3.08% count error. Median per-coil error falls from 46.2% to 44.3%. Neighbouring corrections reach approximately 51% of the coils; about 39% improve and 13% worsen. These are historic diagnostics with incomplete directional support, not final count-validation estimates.

A 5,000-replicate bootstrap that resamples whole coils, retaining their directions and dates, gives a conditional 95% interval of 0.39–7.02 percentage points for the pooled reduction. Removing each date in turn gives reductions between 2.93 and 3.26 points. These checks describe sensitivity within the archived experiment. They do not account for dependence between nearby coils, the UTC mismatch or model uncertainty. A local-clock repeat is necessary before treating these as final count-validation estimates.

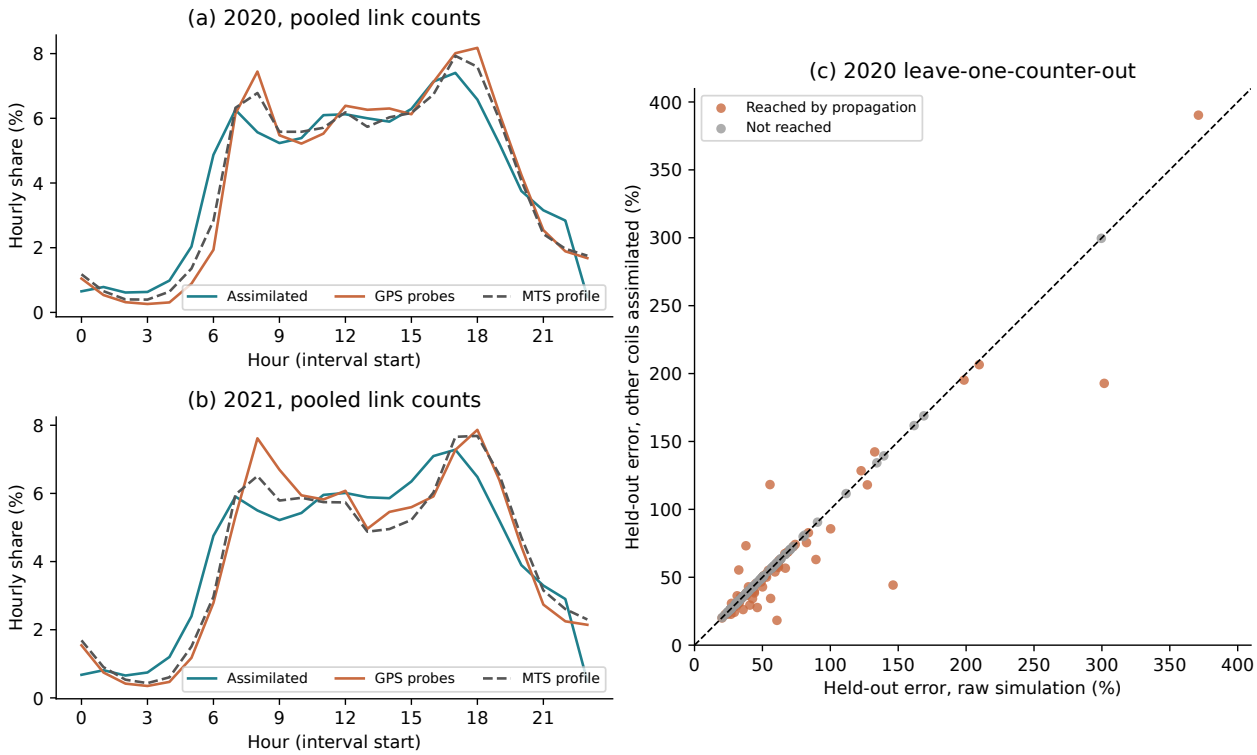


Fig. 6 Timing and withheld-counter diagnostics. (a,b) Pooled assimilated and GPS hourly shares and the MTS baseline for each campaign. Pooling weights links separately by simulated and probe volumes, so these curves do not estimate regional traffic shares. **(c)** Error on each held-out coil before and after assimilation of the others; each point aggregates its available directions and dates. Colour indicates whether neighbouring corrections reach the link. The dashed line denotes equal errors. The counter test retains the archived UTC-as-local convention.

3 Methods

3.1 Demand and network support

The network is derived from OpenStreetMap [46], whose road coverage has been estimated to exceed 95% in most European countries [47]. Its preprocessed polylines store connectivity, lengths, directionality and turn constraints. Graph construction and simplification are distinct from route assignment [48]. Municipality OD activity from January 2020 is averaged separately for Monday–Friday and Saturday–Sunday; public holidays on 1 and 6 January enter the weekday average. The generator represents observed municipality and province pairs explicitly and samples missing endpoints for residual mobility. Activity-to-agent conversion and random endpoints remain modelling assumptions, not observed vehicle identities.

3.2 Simulation and output measurement

The `minimocas` agent-based model advances at 10 s steps and writes hourly outputs. Agents follow destination-dependent best paths computed on the weighted road graph with a Dijkstra-style engine [49]; assignment is not equilibrated in the sense of Wardrop [12], static or dynamic. Link speeds depend on density through the implemented threshold rule. The resulting model is simpler than established simulation platforms [50, 51] and does not resolve the range of congestion phenomena described by microscopic traffic-flow models [52].

Virtual barriers count point crossings; unique-link tables count distinct agents present on a directed polyline in an output interval. The latter can include agents spanning interval boundaries and are not automatically equivalent to physical detector passages. GPS validation sums the two model directions because the supplied comparison uses total link probes. The simulator source initializes its generator with seed 1; repeated runs are not an uncertainty ensemble. A deterministic seed alone does not guarantee bitwise agreement with historical outputs across software and platform versions.

3.3 Archived counter correction

The historic correction code, dated 30 March 2023, maps MTS directions to graph directions. Let $u_i(h)$ and $b_i(h)$ be unique-link and virtual-barrier counts and $c_i(h)$ the mapped counter observation. It computes

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \qquad \Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}, \tag{5}$$

and an additive unique-link increment

$$\delta_i(h) = u_i(h)\Delta_i(h), \qquad \widetilde{u}_i(h) = u_i(h) + \delta_i(h). \tag{6}$$

For $u_i, b_i \geq 1$, this gives $\widetilde{u}_i = c_i$ when the update is applied. Corrections propagate forwards and backwards through adjacent links, using simulated turning shares, with a nominal depth limit of three and a one-vehicle

increment threshold. Forbidden turns, closed links and other detector links constrain propagation. Negative output values are clipped to zero.

The archived recursion has two implementation qualifications: outgoing sibling weights are compounded and the visited-link counter is shared between branches. Thus its nominal depth is not an independent path-length limit, and the update should not be described as a globally conservative reconstruction. The primary results assess that historic implementation. Corrected algorithms must be evaluated as separate experimental variants rather than silently substituted.

3.4 Validation, baselines and missingness

Every campaign uses exactly the dates present in the GPS extract. Counts are summed over those dates before each link is normalized; every comparison includes 24 hourly bins and applies no time shift. Links with zero model or probe total are excluded. Absent GPS link-hour rows are treated as zero in the supplied extract; the archive does not distinguish no passage from no observation. Raw and corrected simulation tables are checked for duplicate direction keys and missing hourly columns.

The OD baseline sums configuration creation rates in each hour. For 2020, the MTS baseline sums contemporaneous counts across the selected local dates, normalizes each complete direction by its total, and averages direction profiles equally. For 2021, daily summer weekday profiles are constructed from the 2022 averages and then averaged over campaign dates. The revised calculation preserves all-missing hourly cells as missing; the inherited calculation converted them to zero. This changes the 2021 median baseline error from 1.2143 to 1.2156 pp without changing the comparison's conclusion. A uniform baseline assigns 1/24 to every hour.

Weekday and weekend analyses repeat normalization and link eligibility within each subset. No independent-link confidence intervals are used because neighbouring links share vehicle trajectories and model inputs. The counter bootstrap samples 149 coil clusters with replacement, recomputes the pooled error reduction, and takes percentile limits (5,000 replicates, NumPy seed 20261027). It resamples existing held-out predictions; it does not refit the simulation or create independent spatial replicates.

4 Discussion

The results distinguish spatial coverage of a model from the accuracy supported by observations. Mobile-phone demand and network loading produce an hourly value for each directed link. Independent GPS profiles test only the temporal distribution on matched roads in a subregion. With simulation-relative scaling they cannot establish absolute volumes elsewhere. The held-out counter experiment provides limited evidence of transfer of local count information, while leaving more than half an observed daily volume in pooled absolute hourly error.

Temporal baselines are essential to interpreting the GPS agreement, as recommended for the validation of traffic simulations [42, 43]. A high hourly correlation can coexist with limited improvement over the input demand and worse accuracy than a mean counter profile; the strength of historical-average profiles is well known in short-term traffic forecasting [44, 45]. The difference is practically relevant: the simulation supplies

spatial allocation and modelled volumes, but its temporal performance alone does not justify the additional model complexity. The small or negative profile benefit of assimilation also shows that fitting local counts and improving out-of-sample temporal shape are different objectives.

The contribution is therefore a transparent assessment of a heterogeneous regional workflow, including its failure modes. It complements established mobile-phone assignment and sensor-fusion studies [9–11, 31, 32] and count-based calibration of agent-based demand [22]; it does not replace probabilistic sensor-free estimation or sparse-sensor methods [38, 39]. No common-data benchmark against those methods has been performed. A stronger methodological claim would require corrected implementation, genuinely held-out sensor groups and explicit demand/parameter sensitivity.

4.1 Future work

January 2020 demand is reused for a summer 2021 campaign with potentially different mobility levels and behaviour; Italian mobility changed sharply under the 2020 COVID-19 restrictions [53, 54]. Two short campaigns cannot establish seasonal stability or transferability to other regions. Weekend comparisons contain fewer probes and only two dates each. Sensor-class associations partly reflect simulated denominators, and the fleet's hourly representativeness remains unknown. The model graph lacks a verified OSM road-class mapping for stratified assessment.

Time handling requires direct attention. The generator drops a late-night demand component and injects selected hourly OD values earlier than a straightforward start-labelled mapping. The original OD hour semantics must be established before a corrected mapping is interpreted as ground truth. Raw counter tables are UTC, while the archived update uses them as local. The historic 2020 local-clock assimilation sensitivity slightly improves GPS profile error, but a full corrected-demand and corrected-clock counter assessment has not yet been completed. Neither the primary estimates nor their bootstrap intervals resolve those biases.

Further methodological work will address directed output support and the correction rules. A pilot identified loss of a held-out direction on a terminal link in the historic correction, requiring a support-preserving implementation before final count validation. A 10 February 2020 simulation replay completed with matching link and hour keys but did not reproduce the archived numeric counts: summed link-hour counts were 33.36 million against 34.33 million. Its discrepancy has not been isolated and is not evidence of a demand correction. Replacing sibling-weight compounding or changing the depth rule also changes the algorithm and needs independent assessment. Priorities are to reconcile clocks and output definitions, establish simulator-version compatibility, repeat held-out validation with corrected inputs, and measure sensitivity to sensor coverage and OD assumptions. Global count uncertainty and network conservation remain unestablished.

A separate planned study will investigate time-dependent traffic exposure and road safety. It will compare accident counts with exposure-normalized temporal patterns, using accident and traffic observations matched in time and at compatible spatial resolutions, and assess how exposure uncertainty affects the resulting indicators. This application forms the focus of the second manuscript; no road-safety result is established by the present reconstruction assessment.

5 Conclusions

A regional reconstruction from mobile-phone OD activity, a road graph and sparse counters can be evaluated without conflating model coverage with observation support. Across two GPS campaigns, reconstructed temporal profiles perform only slightly better than OD timing and less accurately than a mean counter profile. Held-out counters indicate a modest reduction in count error when neighbouring measurements are used, with substantial residual error. These results identify concrete timing and correction issues to resolve before stronger regional accuracy claims are warranted.

Supplementary Material

The supplement records the data layers, run inventory, model variables, archived correction rules, full validation tables, selected-road illustrations and reproduction commands. Accident and operational-platform analyses are outside the scope of this article.

Availability of data and materials

The present working package contains manuscript sources, analysis scripts and aggregate results. Mobile-phone and Viasat datasets are proprietary and are not redistributed. The regional MTS source is https://mobilita.regione.emilia-romagna.it/servizi/flussi-online; road support is derived from OpenStreetMap. **[TODO: Confirm the permitted route for access to restricted data for review and replication, and archive the exact public counter inputs and licences.]**

Code availability

Analysis scripts for this draft are in `article1/analysis/`; the historic scripts and simulator are retained in the parent project. No public release or archival DOI is claimed. **[TODO: Confirm repository visibility, simulator licensing and a distributable reproducibility package before submission.]**

Ethics approval and consent to participate

The inherited manuscript describes operator-provided, anonymized and aggregated mobile-phone counts, with no individual-level records accessed by the authors. **[TODO: Authors must verify the delivered aggregation level, applicable permissions and the final ethics/consent declaration.]**

Competing interests

C.M. and A.F. are employed by Mindicity, which contributed to the modelling framework. **[TODO: Confirm this inherited declaration and the absence or presence of other interests for every author.]**

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Authors' contributions

The existing author list is preserved. **[TODO: Confirm contributions to this new manuscript and the additional analyses; no final-author approval is asserted by this working draft.]**

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